Paballo Pholo

Graduate Assessment 1 - 03/12/2025

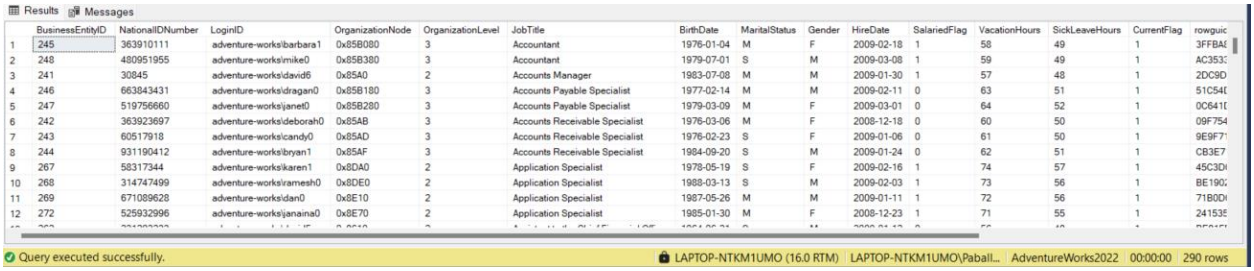
USE [AdventureWorks2022];

1. Write a query in SQL to retrieve all rows and columns from the employee table in the Adventureworks database. Sort the result set in ascending order on jobtitle

SELECT *

FROM [HumanResources].[Employee]

ORDER BY [JobTitle] ASC;



Results Messages

	BusinessEntityID	NationalIDNumber	LoginID	OrganizationNode	OrganizationLevel	JobTitle	BirthDate	MaritalStatus	Gender	HireDate	SalariedFlag	VacationHours	SickLeaveHours	CurrentFlag	rowguic
1	245	363910111	adventure-works\barbara1	0x85B080	3	Accountant	1976-01-04	M	F	2009-02-18	1	58	49	1	3FFBAI
2	248	480951955	adventure-works\mike0	0x85B380	3	Accountant	1979-07-01	S	M	2009-03-08	1	59	49	1	AC353
3	241	30845	adventure-works\david6	0x85A0	2	Accounts Manager	1983-07-08	M	M	2009-01-30	1	57	48	1	2DC9D
4	246	663843431	adventure-works\dragan0	0x85B180	3	Accounts Payable Specialist	1977-02-14	M	M	2009-02-11	0	63	51	1	51C54I
5	247	519756660	adventure-works\janet0	0x85B280	3	Accounts Payable Specialist	1979-03-09	M	F	2009-03-01	0	64	52	1	0C641I
6	242	363923697	adventure-works\deborah0	0x85AB	3	Accounts Receivable Specialist	1976-03-06	M	F	2008-12-18	0	60	50	1	09F754
7	243	60517918	adventure-works\candy0	0x85AD	3	Accounts Receivable Specialist	1976-02-23	S	F	2009-01-06	0	61	50	1	9E9F7
8	244	931190412	adventure-works\bryan1	0x85AF	3	Accounts Receivable Specialist	1984-09-20	S	M	2009-01-24	0	62	51	1	CB3E7
9	267	58317344	adventure-works\karen1	0x8DA0	2	Application Specialist	1978-05-19	S	F	2009-02-16	1	74	57	1	45C3D
10	268	314747499	adventure-works\ramesh0	0x8DE0	2	Application Specialist	1988-03-13	S	M	2009-02-03	1	73	56	1	BE190
11	269	671089628	adventure-works\dan0	0x8E10	2	Application Specialist	1987-05-26	M	M	2009-01-11	1	72	56	1	71B0D
12	272	525932996	adventure-works\janaina0	0x8E70	2	Application Specialist	1985-01-30	M	F	2008-12-23	1	71	55	1	24153

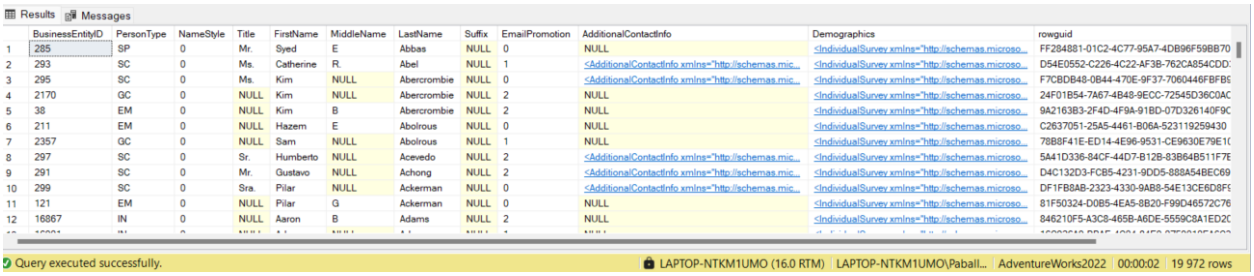
Query executed successfully. LAPTOP-NTKM1UMO (16.0 RTM) LAPTOP-NTKM1UMO\Paball... AdventureWorks2022 00:00:00 290 rows

2. Write a query in SQL to retrieve all rows and columns from the employee table using table aliasing in the Adventureworks database. Sort the output in ascending order on lastname.

SELECT *

FROM [Person].[Person] AS PersonDetails

ORDER BY LastName ASC;



	BusinessEntityID	PersonType	NameStyle	Title	FirstName	MiddleName	LastName	Suffix	EmailPromotion	AdditionalContactInfo	Demographics	rowguid
1	285	SP	0	Mr.	Syed	E	Abbas	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso...	FF284881-01C2-4C77-95A7-4DB96F59BB70
2	293	SC	0	Ms.	Catherine	R.	Abel	NULL	1	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	D54E0552-C226-4C22-AF3B-762CA854CDD3
3	295	SC	0	Ms.	Kim	NULL	Abercrombie	NULL	0	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	F7CBDB48-0B44-470E-9F37-7060446FBFB5
4	2170	GC	0	NULL	Kim	NULL	Abercrombie	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso...	24F01B54-7A67-4B48-9ECC-72545D36C0AC
5	38	EM	0	NULL	Kim	B	Abercrombie	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso...	9A2163B3-2F4D-4F9A-91BD-07D326140F9C
6	211	EM	0	NULL	Hazem	E	Abolrous	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso...	C2637051-25A5-4461-B06A-523119259430
7	2357	GC	0	NULL	Sam	NULL	Abolrous	NULL	1	NULL	<IndividualSurvey xmlns="http://schemas.microso...	78B8F41E-ED14-4E96-9531-CE9630E79E1C
8	297	SC	0	Sr.	Humberto	NULL	Acevedo	NULL	2	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	5A41D336-84CF-44D7-B12B-83B64B511F7E
9	291	SC	0	Mr.	Gustavo	NULL	Achong	NULL	2	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	D4C132D3-FCB5-4231-9DD5-888A54BEC69
10	299	SC	0	Sra.	Pilar	NULL	Ackerman	NULL	0	<AdditionalContactInfo xmlns="http://schemas.mic...	<IndividualSurvey xmlns="http://schemas.microso...	DF1FB8AB-2323-4330-9AB8-54E13CE6D8F5
11	121	EM	0	NULL	Pilar	G	Ackerman	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso...	81F50324-D0B5-4EA5-8B20-F99D46572C76
12	16867	IN	0	NULL	Aaron	B	Adams	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso...	846210F5-A3C8-465B-A6DE-5559C8A1ED2C

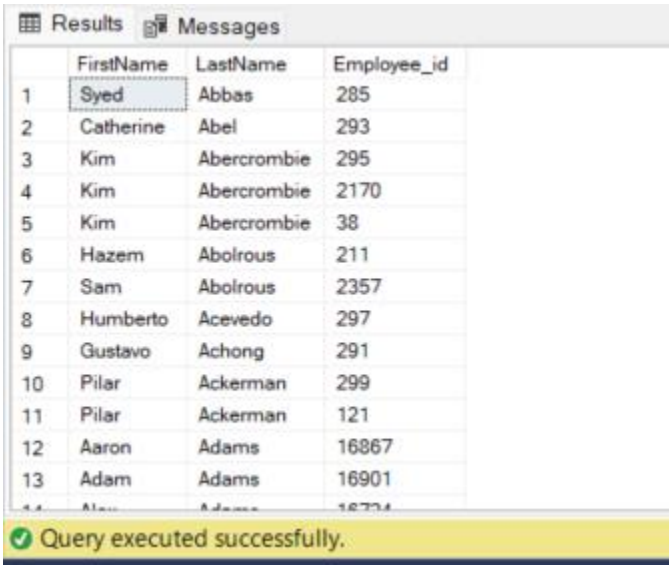
Query executed successfully. LAPTOP-NTKM1UMO (16.0 RTM) LAPTOP-NTKM1UMO\Paball... AdventureWorks2022 00:00:02 19 972 rows

3. Write a query in SQL to return all rows and a subset of the columns (FirstName, LastName, businessentityid) from the person table in the AdventureWorks database. The third column heading is renamed to Employee_id. Arranged the output in ascending order by lastname.

```
SELECT FirstName, LastName, Businessentityid AS Employee_id

FROM [Person].[Person]

ORDER BY LastName ASC;
```

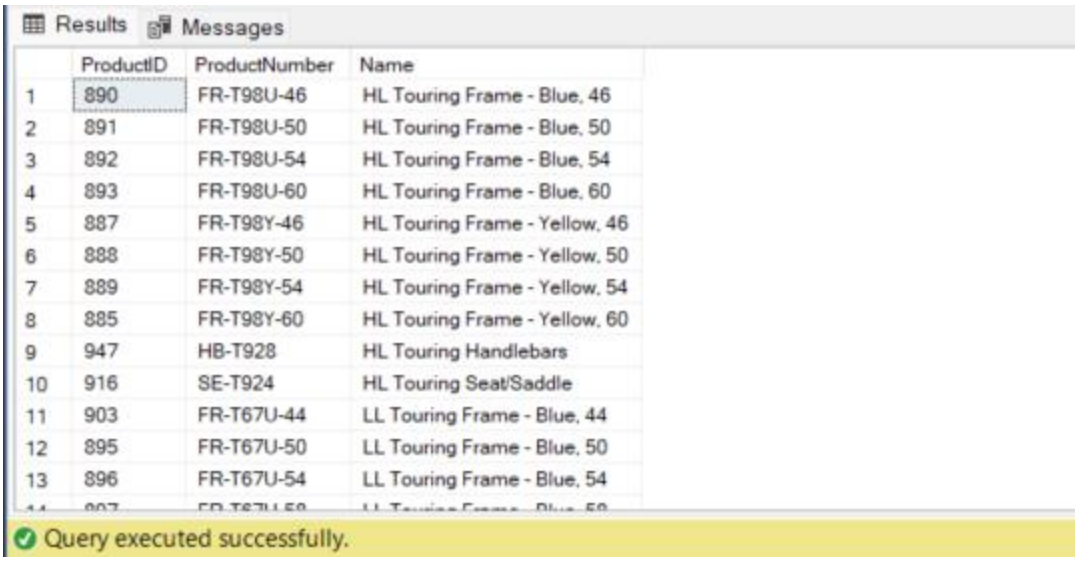


Results Messages

	FirstName	LastName	Employee_id
1	Syed	Abbas	285
2	Catherine	Abel	293
3	Kim	Abercrombie	295
4	Kim	Abercrombie	2170
5	Kim	Abercrombie	38
6	Hazem	Abolrous	211
7	Sam	Abolrous	2357
8	Humberto	Acevedo	297
9	Gustavo	Achong	291
10	Pilar	Ackerman	299
11	Pilar	Ackerman	121
12	Aaron	Adams	16867
13	Adam	Adams	16901

Query executed successfully.

4. Write a query in SQL to return only the rows for product that have a sellstartdate that is not NULL and a productline of 'T'. Return productid, productnumber, and name. Arranged the output in ascending order on name.


Results Messages

	ProductID	ProductNumber	Name
1	890	FR-T98U-46	HL Touring Frame - Blue, 46
2	891	FR-T98U-50	HL Touring Frame - Blue, 50
3	892	FR-T98U-54	HL Touring Frame - Blue, 54
4	893	FR-T98U-60	HL Touring Frame - Blue, 60
5	887	FR-T98Y-46	HL Touring Frame - Yellow, 46
6	888	FR-T98Y-50	HL Touring Frame - Yellow, 50
7	889	FR-T98Y-54	HL Touring Frame - Yellow, 54
8	885	FR-T98Y-60	HL Touring Frame - Yellow, 60
9	947	HB-T928	HL Touring Handlebars
10	916	SE-T924	HL Touring Seat/Saddle
11	903	FR-T67U-44	LL Touring Frame - Blue, 44
12	895	FR-T67U-50	LL Touring Frame - Blue, 50
13	896	FR-T67U-54	LL Touring Frame - Blue, 54

Query executed successfully.

```
SELECT ProductID, ProductNumber, Name

FROM [Production].[Product]

WHERE [SellStartDate] IS NOT NULL AND [ProductLine] = 'T'

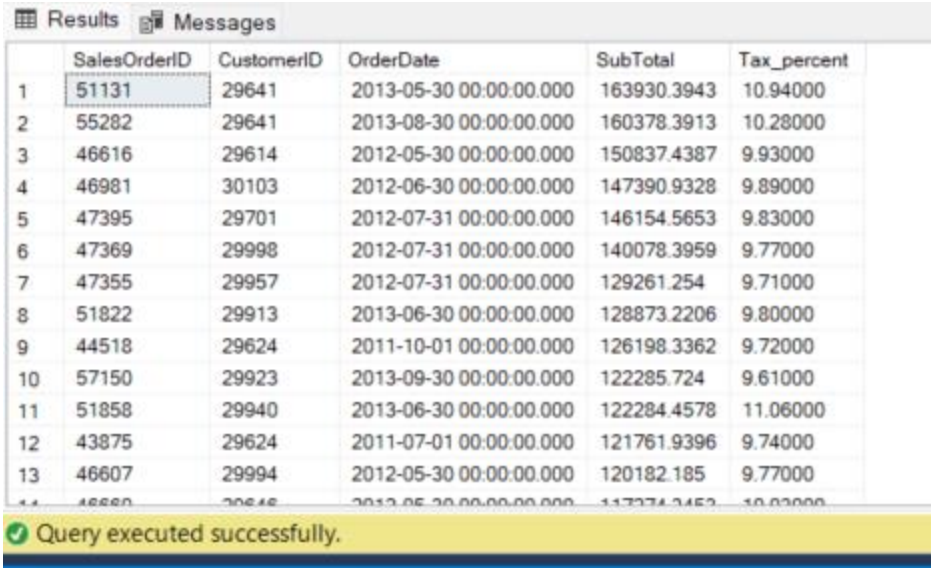
ORDER BY Name ASC;
```

5. Write a query in SQL to return all rows from the salesorderheader table in Adventureworks database and calculate the percentage of tax on the subtotal have decided. Return salesorderid, customerid, orderdate, subtotal, percentage of tax column. Arranged the result set in ascending order on subtotal.

```
SELECT SalesOrderID, CustomerID, OrderDate, SubTotal, ((TaxAmt/SubTotal)*100.0) AS Tax_percent

FROM [Sales].[SalesOrderHeader]

ORDER BY SubTotal DESC;
```



Results Messages

	SalesOrderID	CustomerID	OrderDate	SubTotal	Tax_percent
1	51131	29641	2013-05-30 00:00:00.000	163930.3943	10.94000
2	55282	29641	2013-08-30 00:00:00.000	160378.3913	10.28000
3	46616	29614	2012-05-30 00:00:00.000	150837.4387	9.93000
4	46981	30103	2012-06-30 00:00:00.000	147390.9328	9.89000
5	47395	29701	2012-07-31 00:00:00.000	146154.5653	9.83000
6	47369	29998	2012-07-31 00:00:00.000	140078.3959	9.77000
7	47355	29957	2012-07-31 00:00:00.000	129261.254	9.71000
8	51822	29913	2013-06-30 00:00:00.000	128873.2206	9.80000
9	44518	29624	2011-10-01 00:00:00.000	126198.3362	9.72000
10	57150	29923	2013-09-30 00:00:00.000	122285.724	9.61000
11	51858	29940	2013-06-30 00:00:00.000	122284.4578	11.06000
12	43875	29624	2011-07-01 00:00:00.000	121761.9396	9.74000
13	46607	29994	2012-05-30 00:00:00.000	120182.185	9.77000

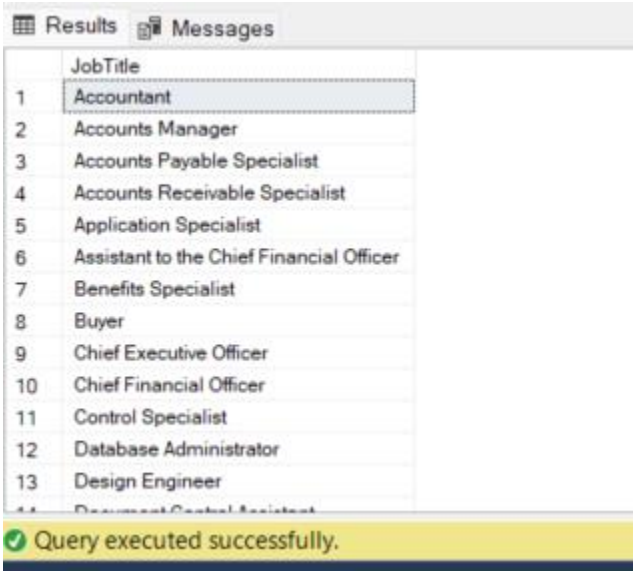
Query executed successfully.

6. Write a query in SQL to create a list of unique jobtitles in the employee table in Adventureworks database. Return jobtitle column and arranged the resultset in ascending order.

```
SELECT DISTINCT [JobTitle]

FROM [HumanResources].[Employee]

ORDER BY [JobTitle] ASC;
```



Results Messages

	JobTitle
1	Accountant
2	Accounts Manager
3	Accounts Payable Specialist
4	Accounts Receivable Specialist
5	Application Specialist
6	Assistant to the Chief Financial Officer
7	Benefits Specialist
8	Buyer
9	Chief Executive Officer
10	Chief Financial Officer
11	Control Specialist
12	Database Administrator
13	Design Engineer

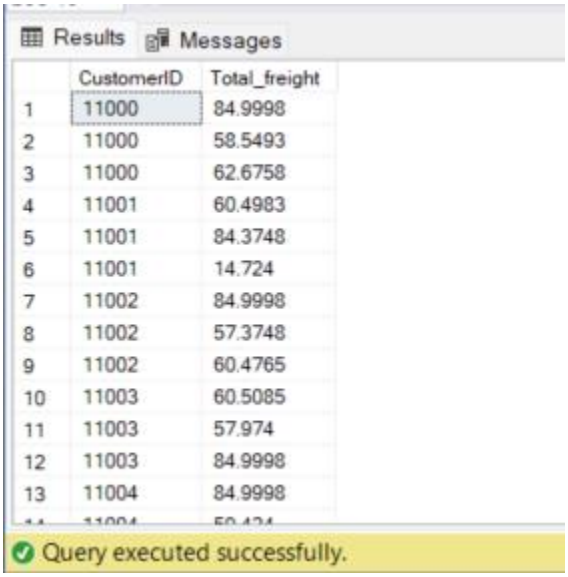
Query executed successfully.

7. Write a query in SQL to calculate the total freight paid by each customer. Return customerid and total freight. Sort the output in ascending order on customerid.

```
SELECT [CustomerID], [Freight] AS Total_freight

FROM [Sales].[SalesOrderHeader]

ORDER BY [CustomerID] ASC;
```



Results Messages

	CustomerID	Total_freight
1	11000	84.9998
2	11000	58.5493
3	11000	62.6758
4	11001	60.4983
5	11001	84.3748
6	11001	14.724
7	11002	84.9998
8	11002	57.3748
9	11002	60.4765
10	11003	60.5085
11	11003	57.974
12	11003	84.9998
13	11004	84.9998

Query executed successfully.

8. Write a query in SQL to find the average and the sum of the subtotal for every customer. Return customerid, average and sum of the subtotal. Grouped the result on customerid and salespersonid. Sort the result on customerid column in descending order

```
SELECT CustomerID, SalesPersonID, AVG(SubTotal) AS avg_subtotal, SUM(SubTotal) AS sum_subtotal

FROM [Sales].[SalesOrderHeader]

GROUP BY [CustomerID], [SalesPersonID]

ORDER BY CustomerID DESC;
```



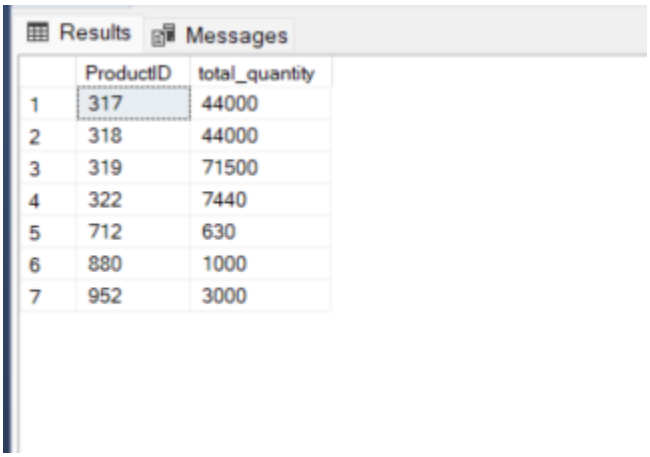
Results Messages

	CustomerID	SalesPersonID	avg_subtotal	sum_subtotal
1	30118	275	34638.1521	207828.9131
2	30118	277	35369.8284	70739.6569
3	30117	275	77171.7928	463030.757
4	30117	277	58954.1365	353724.8193
5	30116	276	46778.5502	187114.2011
6	30115	289	1114.6949	8917.5594
7	30114	290	1456.6238	11652.9911
8	30113	282	34148.2363	273185.8908
9	30112	280	93591.6217	93591.6217
10	30112	284	54909.3785	384365.6498
11	30111	277	69131.6007	276526.4031
12	30110	276	406.3188	1625.2755
13	30109	275	59857.2605	119714.521

Query executed successfully.

9. Write a query in SQL to retrieve total quantity of each productid which are in shelf of 'A' or 'C' or 'H'. Filter the results for sum quantity is more than 500. Return productid and sum of the quantity. Sort the results according to the productid in ascending order

```
SELECT p.[ProductID], SUM(o.[OrderQty]) AS total_quantity --, p.ProductNumber
FROM [Production].[Product] AS p
JOIN [Purchasing].[PurchaseOrderDetail] AS o
on p.ProductID = o.ProductID
WHERE LEFT(p.ProductNumber,1) LIKE '[ACH]%'
GROUP BY p.[ProductID] --,p.ProductNumber
HAVING SUM(o.OrderQty) > 500
ORDER BY ProductID ASC;
```



Results | Messages

	ProductID	total_quantity
1	317	44000
2	318	44000
3	319	71500
4	322	7440
5	712	630
6	880	1000
7	952	3000

10. Write a query in SQL to find the total quentity for a group of locationid multiplied by 10

```
SELECT *

FROM [Production].[Product] AS p

JOIN [Purchasing].[PurchaseOrderDetail] AS o

on p.ProductID = o.ProductID;
```